

# ABDUL SAMAD

Full-Stack Developer (MERN) | AI/ML Engineering

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## Summary

Final-year B.Tech (AI ML) student with hands-on experience in MERN stack development, backend APIs, and ML-powered applications. Built full-stack systems integrating Flask microservices, scikit-learn models, and Google Gemini APIs. Seeking Software Development, Backend, or AI/ML internship and full-time opportunities.

## Technical Skills

|                               |                                                                                                                                                                  |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Languages:</b>             | Python, JavaScript (ES6+), Java, SQL                                                                                                                             |
| <b>Frontend:</b>              | React, HTML5, CSS3, Tailwind CSS                                                                                                                                 |
| <b>Backend:</b>               | Node.js, Express.js, Flask, REST API Design, Microservices Architecture                                                                                          |
| <b>Databases &amp; Cloud:</b> | MongoDB (Atlas), Mongoose, IBM Cloudant (NoSQL), IBM Cloud, Vercel, Render                                                                                       |
| <b>AI / ML:</b>               | Google Gemini API, Google Generative AI SDK, Prompt Engineering, scikit-learn, Logistic Regression, Random Forest, Naive Bayes, TF-IDF, NLP, Feature Engineering |
| <b>Tools:</b>                 | Git, GitHub, Docker, Postman/Thunder Client, Mapbox GL JS, Recharts                                                                                              |
| <b>Core CS:</b>               | Data Structures & Algorithms, OOP, Operating Systems, DBMS                                                                                                       |

## Projects

### EduPulse AI — AI-Powered Skill Tracking & Procrastination Detection Platform [GitHub](#) | 2026

- Built a full-stack **MERN** application (React, Node.js/Express, MongoDB) with **Mongoose** schema design for skills, tasks, and per-user progress tracking
- Applied **prompt engineering** with the **Google Generative AI SDK** (@google/generative-ai, gemini-2.5-flash) to auto-generate 7–10 structured, difficulty-tagged learning tasks per skill on demand
- Designed a **synthetic dataset** (10,000 rows, 11 engineered features) and trained a **Logistic Regression** procrastination-risk classifier (~82% accuracy, selected over Random Forest); deployed as an independent **Flask** microservice under a **microservices architecture**, integrated with the Node.js backend over internal REST calls
- Implemented a per-skill streak-tracking engine with dual-trigger deadline evaluation and persistent completion history, following an additive-only integration pattern to preserve existing functionality
- Tech:** React, Node.js, Express, MongoDB, Mongoose, Flask, scikit-learn, Google Generative AI SDK, Tailwind CSS

### Cloud-Powered Task Manager — Cloud-Native CRUD Task Management App [Live](#) | [GitHub](#) | Jul 2026

- Built a full-stack **CRUD** application (Node.js, Express) storing every task as a JSON document in **IBM Cloudant**, a managed NoSQL database on IBM Cloud, with an automatic in-memory fallback for offline development
- Designed an **idempotent bulk-seed API** using Cloudant's **postBulkDocs** endpoint, resolving a document-conflict bug caused by tombstoned deleted document IDs through revision-aware upsert logic
- Containerized** the application with **Docker** and Docker Compose for consistent, portable deployment; deployed live on **Render** with continuous deployment from GitHub
- Implemented a real-time connection-status API endpoint surfacing live database state, document counts, and storage backend directly in the UI
- Tech:** Node.js, Express, IBM Cloudant, Docker, Render, JavaScript

### Nagar Nazar — Smart City Infrastructure Monitoring Platform [Live](#) | [GitHub](#) | Apr 2026

- Built a real-time **GIS dashboard** (Mapbox GL JS) for monitoring water, sewage, and electric utility infrastructure across city zones
- Integrated a machine learning model to flag high-risk zones from utility data; placed in the **Top 40 of 200+ teams** at HackDiwas 3.0
- Tech:** React, Mapbox GL JS, Machine Learning, REST APIs

### SMS Spam Detection System [Live](#) | [GitHub](#) | Aug 2025

- Trained a **Naive Bayes** classifier with **TF-IDF** feature extraction, achieving **97% accuracy** on message classification
- Deployed the model as a **Flask REST API** on Vercel with a lightweight web front end for live predictions
- Tech:** Python, scikit-learn, NLP, TF-IDF, Flask, JavaScript, Vercel

## Education

**United Institute of Technology**, Prayagraj 2023 – 2027

B.Tech in Computer Science & Engineering | Specialization: AI & ML | CGPA: **7.84**

**Allahabad Public School & College**, Prayagraj

Senior Secondary (CBSE), 2022–2023 | Secondary (CBSE), 2020–2021

## Certifications & Training

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional** — Oracle (2025)
- AWS Solutions Architecture Job Simulation** — AWS (2025)
- Google Cloud Skills Boost** — GCP Fundamentals (2024)
- Generative AI using NLP** — United Global Infoservice Pvt. Ltd. (Aug 2025): tokenization, stemming, TF-IDF; capstone project on SMS spam detection
- Python with Data Science** — United Group of Institutions (Aug 2024): Python programming, data handling, Tkinter GUI development
- Hackathon Achievements:** HackDiwas 3.0 — Top 40 of 200+ teams | UHack 4.0 — Top 65 of 250+ teams | AWS Student Community Day, MNNIT Prayagraj (Apr 2026)